Lecture 9

Def: A **diagonal matrix** is a **square matrix** where all of the non-diagonal entries are zero

Def: A **zero matrix** denoted by 0 is a matrix where every entry is zero

We can only add the matrix with the same size.
We simply add their entries

$$(A+B)_{ij} = A_{ij} + B_{ij}$$

Multiply a matrix by a scalar:
We multiply every entry by that scalar

$$(cA)_{ij} = cA_{ij}$$

Properties: $A, B, C \in \mathbb{R}^{m \times n}$; $r, s \in \mathbb{R}$

$$\begin{cases} (A+B)+C = A+(B+C) \\ A+B = B+A \\ A+0 = A = 0+A \end{cases}$$

$$r(A+B) = rA + rB$$

$$(r+s)A = rA + sA$$

$$r(sA) = (rs)A$$

Multiplication of matrix:

$B \in \mathbb{R}^{m \times n}$, $\vec{x} \in \mathbb{R}^n$

$B\vec{x} \in \mathbb{R}^m$

$A \in \mathbb{R}^{p \times m}$

$A(B\vec{x}) \in \mathbb{R}^p$

property: $A(B\vec{x}) = (AB)\vec{x}$

$$(AB)C = A(BC)$$

$$A(B+C) = AB + AC$$

$$(B+C)A = BA + CA$$

$$I_m A = A = A I_n$$

$$r(AB) = (rA)B = A(rB)$$

Warning! In general:

$AB \neq BA$

$AB = AC \not\Rightarrow B = C$

$AB = 0 \not\Rightarrow A = 0$ or $B = 0$

Def: A is a square matrix and $k = 1, 2, 3, \cdots$ is a positive integer

$$A^k = \underbrace{AA \cdots A}_{k \text{ times}}$$

Def: $A \in \mathbb{R}^{m \times n}$

$A^T \in \mathbb{R}^{n \times m}$

$A^T_{ij} = A_{ji}$

Properties:

$(A^T)^T = A$

$(A+B)^T = A^T + B^T$

$rA^T = (rA)^T$ for any scalar $r \in \mathbb{R}$

$(AB)^T = B^T A^T$